CAPSTONE PROJECT - I REVIEW 1 PRESENTATION

Real-Time Financial Fraud & Credit Risk Intelligence System

End-to-End MLOps Framework with Apache Kafka, GNNs, Concept Drift & Explainable AI

MENTOR & TEAM DETAILS

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PROJECT BACKGROUND & BOTTLENECKS
Problem Statement & Core Industry Challenges

1. Massive Financial Losses (₹3.3 Lakh Crore)

Digital payment fraud causes over **₹3.3 Lakh Crore ($40B+) in annual losses worldwide**, severely threatening high-volume Indian UPI, IMPS, and banking gateways with mounting operational risk.

2. Vulnerability of Legacy Rule Engines

Traditional banking security relies on manual static rules that fail to detect novel zero-day fraud tactics, adaptive criminal behaviors, and dynamic multi-account collusive attacks.

3. Severe ~3.5% Fraud Data Imbalance

Because genuine transactions dominate (~96.5%), standard AI models suffer from severe majority-class bias, missing real fraud or causing high false-positive rejections for legitimate users.

4. Lack of Regulatory Transparency

Black-box deep learning models fail to provide clear, human-understandable explanations for credit or loan rejections, violating strict RBI, FCRA, and GDPR regulatory compliance laws.

THE STORY THAT INSPIRED OUR PROJECT
Real-World Motivation

Multi-Crore Financial Impact

Indian digital payment networks (UPI, IMPS, Cards) process billions of transactions monthly. Undetected fraud breaches cause direct multi-Crore losses to banks and destroy customer trust.

Collusive Fraud Rings

Organized criminal syndicates operate complex inter-account networks to launder illicit funds across banks. Standard tabular classifiers fail to identify these multi-node graph relationships.

RBI Compliance & Audit Trails

Financial regulators strictly require explainable feature attributions (SHAP values) whenever loan requests or credit card applications are flagged or rejected to ensure complete audit compliance.

Sub-100ms Live Execution

High-speed payment gateways demand instantaneous real-time inference under 100 milliseconds to block fraudulent payment attempts before transaction finalization.

PROJECT SCOPE & TARGETS

Objectives and Industry Applications

End-to-End MLOps Core Objectives

1. Sub-100ms Inference API Gateway

Architect high-throughput FastAPI REST microservices serving real-time risk predictions for high-volume banking payment gateways.

2. Big Data Streaming (Apache Kafka)

Stream live transaction data to simulate real-world high-throughput banking pipelines and trigger real-time ML inference.

3. Adaptive Intelligence (Concept Drift)

Implement automated data drift monitoring to detect evolving fraud tactics and trigger dynamic model retraining.

4. Regulatory Explainability & Audit Logs (SHAP)

Compute instant Shapley feature attributions to satisfy RBI, FCRA, and GDPR transparency & auditability requirements.

Industry Applications & Deployment

1. Real-Time UPI & Credit Card Interception

Instant transaction risk scoring prior to payment settlement on merchant gateways & card processing networks.

2. Automated Digital Lending & Credit Risk (INR ₹)

Dynamic credit risk scoring (0–100 scale) for personal loan approval automation & default prevention.

3. Enterprise Anti-Money Laundering (AML)

Continuous graph-based fraud ring monitoring for high-frequency financial institutions and regulatory agencies.

4. Core Banking & NetBanking Protection

Real-time anomaly filtering for high-value IMPS, NEFT, and RTGS electronic fund transfers.

PRIOR ACADEMIC STUDIES COMPARISON

Literature Review & Research Gap

Paper	Approach	Strength	Limitation
Danda et al., 2026 – Review	Identifies Big Data & Stream Processing as future	Highlights the need for real-time Kafka architecture	Lacks applied ML implementation
Chen et al., 2025 – Review	Identifies GNNs and XAI as the critical path	Highlights need for regulatory compliance	Only a theoretical framework
Veličković et al., 2018 – GAT	Self-attention over node graph structures	Mines multi-account collusive fraud rings	Requires graph construction pipeline overhead
Bifet et al., 2007 – ADWIN	Concept Drift Detection for streaming data	Automatically detects shifting fraud patterns	Requires continuous data streaming pipeline

Research Gap Identified:

• No prior applied study unifies **Apache Kafka Streaming, Concept Drift Adaptation, GNNs, and XAI** in a single containerized MLOps pipeline. Applied papers still rely on static batch processing (CSV) and basic tabular ML.

END-TO-END DEEP LEARNING PIPELINE
Methodology & System Architecture Flowchart

STAGE 01

Data Ingestion

Real-Time Transaction Streaming via Apache Kafka

Kafka Cluster

STAGE 02

Adaptive Intelligence

ADWIN Concept Drift Detection & Model Retraining

Drift Detection

STAGE 03

Tri-Model Fusion Engine

LightGBM + Autoencoder + PyTorch GAT

Deep & Graph Models

STAGE 04

Risk Synthesis

Weighted Score Synthesis & SHAP Explainability Engine

SHAP Output

STAGE 05

FastAPI Gateway

Docker Containerized FastAPI & MLflow Tracking

MLOps / Docker

5 Enterprise Layers (Kafka, Fusion, MLOps)

< 100ms Target Real-Time Inference Latency

100% RBI & FCRA Regulatory Audit Compliance

CORE INNOVATION PILLARS
Novelty of the Idea – What Makes Our Project Unique

Hybrid Tri-Model Fusion Engine

Deploys a complete enterprise architecture using **Apache Kafka**, **MLflow**, and **Docker**, moving beyond static student notebook projects.

Tri-Model Core Architecture

Fuses **LightGBM** (Tabular), **Autoencoder** (Zero-day fraud), and **GAT** (Fraud Rings) for comprehensive threat detection.

Explainable AI & Audit Logging (SHAP)

Provides instant **Shapley feature attributions** for full **RBI, FCRA & GDPR compliance** on flagged and rejected loan or transaction attempts.

Production MLOps & Drift Monitoring

Continuous automated data drift & concept decay monitoring powered by **Evidently AI** and **FastAPI microservice integration**.

CAPSTONE MILESTONES & TIMELINE
Project Implementation Roadmap & Next Steps

Phase 1: Foundation & Initial Review (Weeks 1–4)

✔ COMPLETED

- **Dataset & Problem Setup:** Acquired Kaggle / IEEE-CIS financial transactions & constructed transaction graph
- **Baseline Prototyping:** Evaluated initial LightGBM classifier & Autoencoder anomaly detector
- **Review Submission:** Completed comprehensive literature review & Review-1 defense documentation

Phase 2: Model Training & Feature Integration (Weeks 5–8)

⚡ NEXT STEPS / IN PROGRESS

- **Streaming & Tracking:** Implement Apache Kafka streaming & MLflow experiment tracking
- **Deep Neural & Graph Models:** Train Autoencoder for zero-day fraud & PyTorch Graph Attention Networks (GAT)
- **Explainable AI (XAI):** Integrate SHAP Shapley values to generate regulatory decision audit logs

Phase 3: System Deployment & Final Evaluation (Weeks 9–12)

🎯 TARGET GOAL

- **FastAPI Backend Gateway:** Build real-time scoring microservice (<100ms response time)
- **Live Interactive Dashboard:** Streamlit frontend for real-time fraud monitoring & credit risk simulation
- **System Benchmarking:** Conduct stress testing, data drift evaluation & final Review-2 presentation

CITATIONS & FRAMEWORKS
Academic References & Technical Tools

Academic References:

1. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*. NeurIPS 2017.

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3. Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). *Graph Attention Networks*. ICLR (2018).

4. Xu, L., Skoularidou, M., Cuesta-Infante, A., & Veeramachaneni, K. (2019). *Modeling Tabular Data using Drift Detection*. NeurIPS (2019).

Thank You!

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PROJECT TEAM & FACULTY MENTOR ACKNOWLEDGMENT

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